

-From Predicting Frames to Explaining Movies: *Evaluating Causality, Affordances, Beliefs, and Counterfactuals* in **World Models**

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Abstract

World models are increasingly viewed as foundations for agents, robotics, and physical AI, yet their evaluation remains dominated by *future-state prediction*: reconstruction quality, latent retrieval, temporal consistency, planning success, and rollout fidelity. These metrics test whether a model can forecast observations, but not whether it understands the dependency structure that generates them. A model may predict what happens next without knowing how scenes progress, how actions unfold, which events matter, or why future outcomes become possible.

We introduce **EXPLAINWORLD**, a benchmark for measuring world understanding beyond prediction. Built upon MovieNet, EXPLAINWORLD treats movies as temporally extended, agent-rich, partially observable worlds and evaluates whether learned world representations capture explanatory structure rather than predictive regularities. To reduce shortcut solutions, tasks are grounded in aligned movie scenes, action annotations, scripts, subtitles, character identities, and place labels, and can be evaluated under video-only, text-only, and multimodal settings.

EXPLAINWORLD is organized around **RIFT**: Reasoning over Interactions, Futures, and Trajectories. RIFT measures world understanding through three complementary dimensions: **Scene Dynamics**, which tests whether models recover next scenes, missing scenes, and long-horizon future scenes; **Action Dynamics**, which tests next-action prediction, missing-action recovery, and action-order verification using MovieNet’s 80 action classes; and **Causal Dynamics**, which tests whether models retrieve causally relevant predecessor scenes and remain sensitive to removing pivotal events.

Using MovieNet’s aligned movies, scripts, subtitles, synopses, character identities, scene boundaries, action labels, and place annotations, we construct retrieval, classification, ranking, and perturbation tasks that probe world understanding at scene, action, and causal levels. This design evaluates whether models capture **what happens next**, **what action should occur**, and **which prior events make future outcomes possible**.

We evaluate representative world-model families, including latent predictive world models (**V-JEPA**, **V-JEPA 2**), action-conditioned world models (**DreamerV3**-style latent dynamics models), interactive world models (**Genie**-style controllable simulators), and world foundation models (**Cosmos**-style physical-world simulators).

We define the **World Understanding Gap (WUG)** as

$$\text{WUG}_m = \left[1 - \frac{U_m}{P_m + \epsilon} \right]_+,$$

where P_m is normalized predictive performance and U_m is the aggregate RIFT understanding score over Scene, Action, and Causal Dynamics. WUG quantifies the fraction of predictive competence not matched by explanatory world understanding.

Across model families, EXPLAINWORLD is designed to reveal whether predictive competence is accompanied by explanatory competence. In particular, it tests whether models that forecast future scenes also recover missing events, preserve action order, retrieve causal predecessors, and respond appropriately when pivotal events are removed. **These evaluations make predictive accuracy an explicit object of scrutiny rather than an assumed proxy for world understanding.** EXPLAINWORLD provides a benchmark, framework, and diagnostic metric for measuring whether world models merely predict future observations or understand the scene, action, and causal dependencies that generate them.

1 Do World Models Understand the Worlds They Predict?

World models are increasingly viewed as foundations for agents, robotics, and physical AI. From latent predictive models and action-conditioned simulators to generative world models, recent

progress shares a common objective: learning representations that anticipate how the world evolves. Consequently, evaluation has largely centered on future-state prediction, reconstruction quality, retrieval accuracy, temporal consistency, and rollout fidelity. Yet these metrics answer only one question: *can a model forecast future observations?* They do not answer the deeper question: **does the model understand the world that generates those observations?** A model may correctly predict what happens next while failing to explain why it happened, what actions were possible, what agents knew or falsely believed, or how events would unfold under alternative interventions. This motivates the central question of this paper: **do world models understand the worlds they predict, or do they merely forecast their future observations?**

1.1 Contemporary World Models are Simulation Engines

Recent progress in world modeling reflects a shift from *representation learning* toward **simulation learning**. At their core, world models learn compact latent dynamics that allow agents to anticipate future states, evaluate hypothetical actions, and reason over imagined futures (Ha and Schmidhuber, 2018). This view aligns with model-based reinforcement learning, where learned dynamics are rolled forward under candidate actions for planning and control (Hafner et al., 2023). Thus, a world model is not merely an encoder of the present; it is a predictive dynamical system for simulating how an environment evolves.

Contemporary **world-model families** instantiate this simulation view in different ways. **Latent predictive world models** such as V-JEPA and V-JEPA 2 predict future latent states rather than reconstruct pixels (Bardes et al., 2024; Assran et al., 2025). **Action-conditioned world models** such as Dreamer roll latent dynamics forward under hypothetical actions (Hafner et al., 2023). **Interactive world models** such as Genie learn action-controllable environments from unlabeled video (Bruce et al., 2024). **World foundation models** such as Cosmos position learned simulators as reusable infrastructure for robotics and physical AI (Agarwal et al., 2025).

Operationally, this simulation view is reflected in evaluation probes. **Future prediction** tests whether a model anticipates upcoming states; **temporal consistency** tests whether representations preserve event progression; **latent retrieval** tests whether predicted representations retrieve the correct future; **planning or control success** tests whether imagined futures support action selection; and **rollout fidelity** tests whether generated trajectories remain coherent over time (Bardes et al., 2024; Assran et al., 2025; Hafner et al., 2023). These probes ask whether a representation behaves like a simulator by preserving semantics, dynamics, temporal order, action effects, robustness, and open-loop coherence. However, they still measure **simulated continuation**, not **explanatory understanding**: whether the model understands why an event occurred, what actions were possible, what agents believed, or how outcomes would change under intervention.

Moving from simulation to understanding therefore requires a different evaluative target. Instead of asking only whether a model can continue a trajectory, we must ask whether it can explain the structure that makes that trajectory possible: its causes, constraints, hidden agent states, and alternatives.

1.2 The Evaluation Gap: Prediction $\neq$ Understanding

Existing world-model evaluations primarily ask whether models can **continue the world**: predict future states, retrieve future clips, support planning, or produce coherent rollouts. Adjacent video-reasoning benchmarks probe narrower explanatory skills: CLEVRER tests causal and temporal reasoning (Yi et al., 2020), CausalVQA tests causal, hypothetical, and planning questions (Foss et al., 2025), COVER targets counterfactual reasoning (Zhou et al., 2025), and VisPhyWorld probes executable physical understanding beyond recognition-based evaluation (Liang et al., 2026). These benchmarks are valuable, but they do not measure **explanatory world understanding** as a unified capability.

This leaves two gaps. First, causal reasoning, affordance understanding, belief tracking, and counterfactual reasoning are typically evaluated in isolation, so success on one axis provides limited evidence about the others. Second, most benchmarks do not test whether stronger prediction implies stronger understanding; they do not measure the gap between **predictive competence** and **explanatory competence**.

Consequently, we lack a unified evaluation of whether world models understand the latent structure of the worlds they predict: the **mechanisms** that cause events, **interactions** that constrain actions, **beliefs** that shape agent behavior, and **alternative futures** that follow from intervention.

2 EXPLAINWORLD: Measuring World Understanding

World understanding requires more than forecasting future observations. A model may predict the next scene while failing to represent the dependency structure that makes that scene possible: how prior events constrain later scenes, how actions unfold over time, and which events causally shape future outcomes. We introduce **EXPLAINWORLD**, a benchmark for measuring this gap between predictive continuation and explanatory world understanding.

EXPLAINWORLD is organized around **RIFT: Reasoning over Interactions, Futures, and Trajectories**. RIFT evaluates whether world models capture three forms of structure underlying world evolution: **Scene Dynamics** for scene-level temporal progression, **Action Dynamics** for ordered action evolution, and **Causal Dynamics** for events that drive downstream outcomes.

We instantiate EXPLAINWORLD using MovieNet, whose aligned movies, scene boundaries, scripts, subtitles, character identities, action annotations, and place labels provide a natural substrate for constructing world-model capability tests. Rather than treating movies as generic video clips, EXPLAINWORLD treats them as temporally structured worlds and evaluates whether models recover the scene, action, and causal dependencies that govern their evolution.

2.1 Scene Dynamics

Scene Dynamics evaluates whether a world model captures the temporal evolution of events at the scene level. Unlike conventional future-prediction evaluations that focus on visual continuation, Scene Dynamics measures whether a model can recover coherent scene transitions, infer missing narrative events, and maintain consistency over extended temporal horizons. All tasks are formulated as retrieval problems over temporally ordered MovieNet scene sequences.

2.1.1 Next-Scene Retrieval

A world model that understands a scene should be able to identify what happens next. Given a query scene S_t , the model retrieves the most plausible successor scene from a candidate pool containing the ground-truth next scene and distractors.

Formally, let

$$\mathcal{C} = \{S_{t+1}, D_1, \dots, D_K\}$$

denote the candidate set, where S_{t+1} is the true successor scene and D_i are distractor scenes. The model computes a compatibility score

$$s(S_t, S_j)$$

for each candidate scene $S_j \in \mathcal{C}$ and ranks candidates in descending order.

Example 1 (Titanic).

S_t : *Rose finds an axe in a flooded corridor*

Candidates:

$\{\textit{Rose reaches Jack's cabin, Passengers board lifeboats, Captain inspects the bridge, Cal searches the deck}\}.$

The correct retrieval is

$$S_{t+1} = \textit{Rose reaches Jack's cabin}.$$

Example 2 (Titanic).

S_t : *Lookouts spot an iceberg ahead*

Candidates:

$\{\textit{Warning relayed to the bridge, Jack sketches Rose, Passengers dine, Engine room inspection}\}.$

The correct retrieval is

$$S_{t+1} = \text{Warning relayed to the bridge.}$$

This task evaluates **local temporal understanding** and tests whether a model captures immediate scene-to-scene progression rather than relying on visual similarity alone.

Metric. Performance is measured using Recall@K and Mean Reciprocal Rank (MRR):

$$\text{Recall@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\text{rank}(S_i^+) \leq K),$$

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}(S_i^+)},$$

where S_i^+ denotes the ground-truth successor scene.

Next-Scene Retrieval

Given a query scene S_t , retrieve the correct successor scene S_{t+1} from a candidate pool.

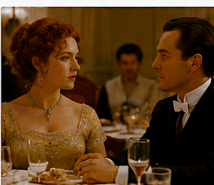
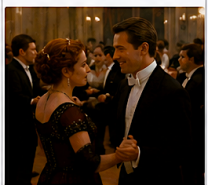
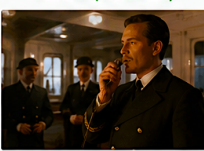
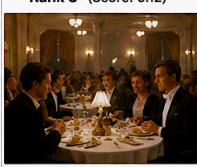
Example 1 (Titanic)		Candidate Pool (including the true next scene)			
Query Scene S_t  Jack wins a ticket in a poker game.	Rank 1 (Score: 0.87)  ✓ Correct Successor Jack boards the Titanic through the gangway.	Rank 2 (Score: 0.21)  Rose has dinner with Cal.	Rank 3 (Score: 0.12)  The captain inspects the bridge.	Rank 4 (Score: 0.07)  Passengers dance below deck.	
Example 2 (Titanic) Query Scene S_t  Lookouts spot an iceberg ahead of the Titanic.	Rank 1 (Score: 0.91)  ✓ Correct Successor The lookout warns the officer on the bridge.	Rank 2 (Score: 0.18)  Jack sketches Rose.	Rank 3 (Score: 0.12)  Passengers dine in the dining room.	Rank 4 (Score: 0.06)  Engine room crew checks the boilers.	
Evaluation Metrics: $\text{Recall@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{rank}(S_i^+) \leq K]$ $\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}(S_i^+)}$  Correct Successor  Distractor					

Figure 1: Examples of **Next-Scene Retrieval**. Given a query scene, the model retrieves the correct successor scene from a candidate pool.

2.1.2 Missing-Scene Retrieval

Narrative progression often depends on intermediate events that connect two observations. Given neighboring scenes S_{t-1} and S_{t+1} , the model retrieves the missing intermediate scene S_t .

Formally,

$$S_{t-1}, \quad ?, \quad S_{t+1} \rightarrow S_t.$$

The candidate pool contains the true intermediate scene together with distractors sampled from the same movie and other movies.

Example 1 (Titanic).

$$S_{t-1} = \text{Rose finds an axe}$$

$$S_{t+1} = \text{Jack escapes confinement}$$

Candidates:

$\{Rose\ cuts\ Jack's\ handcuffs, Passengers\ board\ lifeboats, Engine\ room\ flooding, Cal\ searches\ the\ deck\}$.

The correct retrieval is

$$S_t = Rose\ cuts\ Jack's\ handcuffs.$$

Example 2 (Titanic).

$$S_{t-1} = Rose\ runs\ toward\ the\ stern$$

$$S_{t+1} = Jack\ convinces\ Rose\ to\ step\ back$$

Candidates:

$\{Rose\ climbs\ over\ the\ railing, Passengers\ dance\ below\ deck, Captain\ reviews\ charts, Cal\ attends\ dinner\}$.

The correct retrieval is

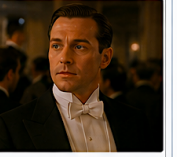
$$S_t = Rose\ climbs\ over\ the\ railing.$$

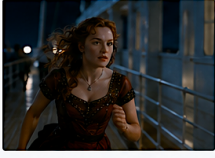
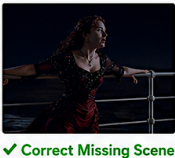
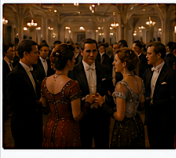
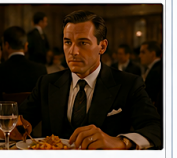
This task evaluates **event coherence** and **temporal interpolation**. Models must recover the scene that best explains the transition between preceding and subsequent events.

Metric. We use Recall@K and MRR, identical to Next-Scene Retrieval.

Missing-Scene Retrieval

Given neighboring scenes S_{t-1} and S_{t+1} , retrieve the missing intermediate scene S_t .

Example 1 (Titanic)		Candidate Pool (including the true missing scene)			
Previous Scene S_{t-1}  Rose finds an axe in a flooded corridor.	Next Scene S_{t+1}  Jack escapes confinement.	Rank 1 (Score: 0.89)  ✓ Correct Missing Scene Rose cuts Jack's handcuffs.	Rank 2 (Score: 0.18)  Passengers board lifeboats.	Rank 3 (Score: 0.07)  Engine room flooding.	Rank 4 (Score: 0.05)  Cal searches the deck.

Example 2 (Titanic)		Candidate Pool (including the true missing scene)			
Previous Scene S_{t-1}  Rose runs toward the stern.	Next Scene S_{t+1}  Jack convinces Rose to step back.	Rank 1 (Score: 0.88)  ✓ Correct Missing Scene Rose climbs over the railing.	Rank 2 (Score: 0.16)  Passengers dance below deck.	Rank 3 (Score: 0.08)  Captain reviews the charts.	Rank 4 (Score: 0.04)  Cal attends dinner.

Evaluation Metrics:	$Recall@K = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{rank}(S_i^+) \leq K]$	$MRR = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}(S_i^+)}$	 Correct Missing Scene	 Distractor
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Note: Scores are example similarity/confidence scores produced by the model. Higher is better.

Figure 2: Examples of **Missing-Scene Retrieval**. The model recovers the missing scene connecting two neighboring observations.

2.1.3 Long-Horizon Scene Retrieval

Local continuity alone does not imply understanding. A world model should also preserve dependencies that span multiple scenes. Given a scene S_t , the model retrieves a future scene S_{t+h} where

$$h \in \{3, 5, 10\}.$$

Formally,

$$S_t \rightarrow S_{t+h}.$$

As the horizon increases, the number of valid intermediate trajectories grows substantially, making retrieval increasingly dependent on narrative and causal structure rather than local visual similarity.

Example 1 (Titanic). Query scene:

$$S_t = \textit{Jack first meets Rose at the stern}$$

Target scene:

$$S_{t+5} = \textit{Jack and Rose dance below deck}.$$

Correct retrieval requires preserving the evolving relationship across multiple intervening scenes.

Example 2 (Titanic). Query scene:

$$S_t = \textit{Lookouts identify an iceberg}$$

Target scene:

$$S_{t+10} = \textit{Passengers evacuate using lifeboats}.$$

Correct retrieval requires maintaining a chain of intermediate events including collision, flooding, warnings, and evacuation procedures.

This task evaluates **long-range temporal consistency** and tests whether representations preserve dependencies beyond immediate continuation.

Metric. Performance is measured using Recall@K and MRR. Results are additionally reported separately for different horizon lengths to quantify degradation as temporal distance increases.

2.2 Action Dynamics

Action Dynamics evaluates whether a world model captures how actions unfold inside temporally segmented movie scenes. While Scene Dynamics operates over scene sequences, Action Dynamics operates over MovieNet’s 80 action classes and tests whether a model understands action progression, missing action dependencies, and temporal action order. This moves evaluation beyond static action recognition toward modeling how actions compose over time.

Let a scene window induce an ordered action sequence

$$A_{1:T} = (a_1, a_2, \dots, a_T), \quad a_t \in \mathcal{A},$$

where $\mathcal{A}$ denotes the MovieNet action vocabulary with $|\mathcal{A}| = 80$. The model assigns a score

$$p_\theta(a \mid \mathcal{H})$$

to each candidate action $a \in \mathcal{A}$, conditioned on the available action context $\mathcal{H}$.

Long-Horizon Scene Retrieval

Given a query scene S_t , retrieve a future scene S_{t+h} where $h \in \{3, 5, 10\}$ from a candidate pool.

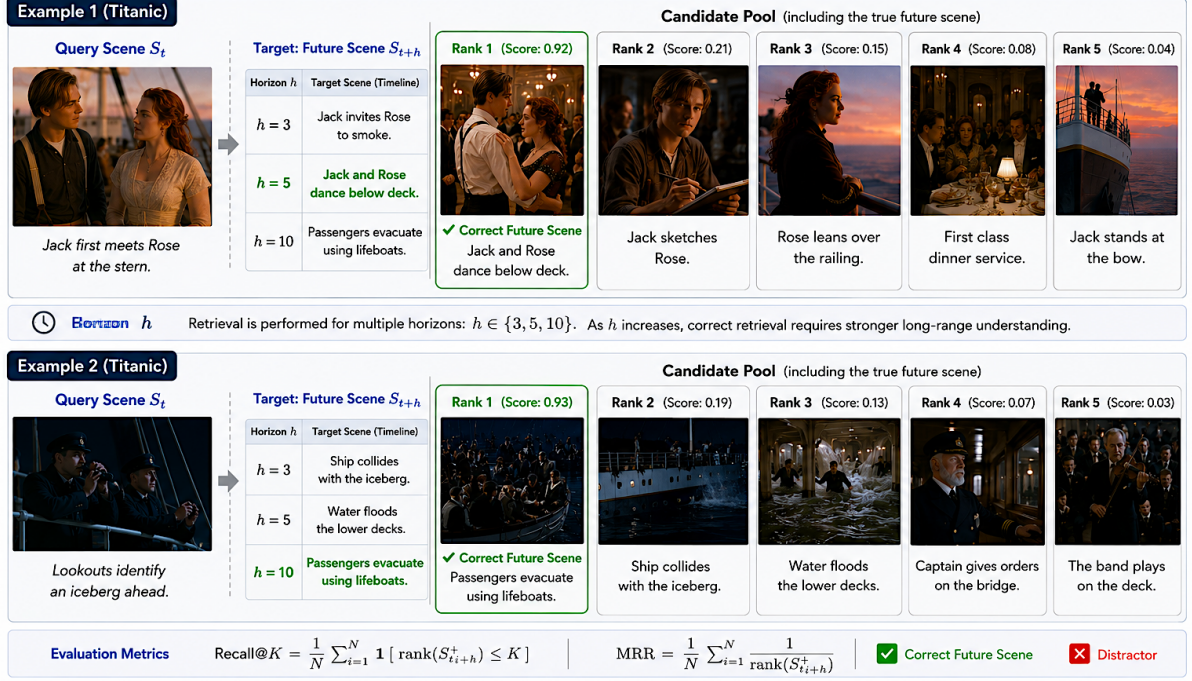


Figure 3: Examples of **Long-Horizon Scene Retrieval**. The model retrieves scenes several steps into the future rather than only the immediate successor.

2.2.1 Next-Action Prediction

Given an action prefix

$$A_{1:t} = (a_1, a_2, \dots, a_t),$$

the model predicts the next action a_{t+1} from the 80-class MovieNet action vocabulary:

$$\hat{a}_{t+1} = \arg \max_{a \in \mathcal{A}} p_{\theta}(a \mid A_{1:t}).$$

This task evaluates whether a model captures **forward action progression**: how ongoing actions constrain what can happen next.

Example 1 (Titanic).

$$A_{1:t} = (\text{run}, \text{reach}, \text{grab axe})$$

Candidate next actions:

$$\{\text{cut}, \text{eat}, \text{dance}, \text{sleep}\}.$$

The correct next action is

$$a_{t+1} = \text{cut}.$$

Example 2 (Titanic).

$$A_{1:t} = (\text{look}, \text{warn}, \text{turn})$$

Candidate next actions:

$$\{\text{collide}, \text{kiss}, \text{sing}, \text{read}\}.$$

The correct next action is

$$a_{t+1} = \text{collide}.$$

Metric. Performance is measured using Top- K accuracy:

$$\text{Acc@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(a_i^+ \in \text{TopK}_{a \in \mathcal{A}} p_{\theta}(a \mid A_{1:t}^{(i)})),$$

where a_i^+ is the ground-truth next action.

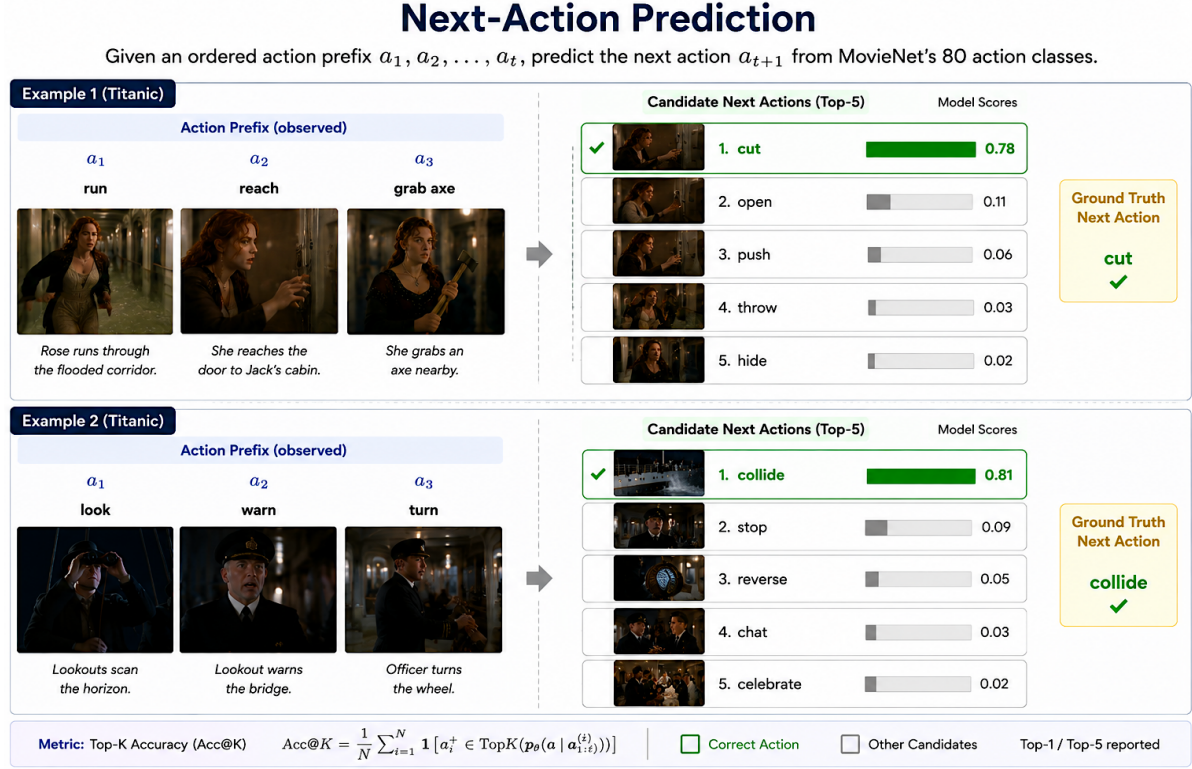


Figure 4: Examples of **Next-Action Prediction**. Given an ordered action prefix, the model predicts the most plausible next action from MovieNet's 80 action classes.

2.2.2 Missing-Action Recovery

Given an action sequence with an intermediate action masked,

$$A_{\setminus t} = (a_1, \dots, a_{t-1}, [\text{MASK}], a_{t+1}, \dots, a_T),$$

the model recovers the missing action a_t :

$$\hat{a}_t = \arg \max_{a \in \mathcal{A}} p_{\theta}(a \mid A_{\setminus t}).$$

This task evaluates **bidirectional action coherence**. Recovering the missing action requires using both preceding and following actions, making the task harder than next-action prediction and less reducible to local action priors.

Example 1 (Titanic).

(find axe, [MASK], escape)

Candidate missing actions:

$\{cut\ handcuffs, eat, play\ guitar, sleep\}$.

The correct missing action is

$a_t = cut\ handcuffs.$

Example 2 (Titanic).

(spot iceberg, [MASK], ship collides)

Candidate missing actions:

$\{\text{warn bridge, dance, kiss, write letter}\}.$

The correct missing action is

$a_t = \text{warn bridge}.$

Metric. Performance is measured using Top- K accuracy:

$$\text{MAR@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(a_i^+ \in \text{TopK}_{a \in \mathcal{A}} p_\theta(a \mid A_{\setminus t}^{(i)})),$$

where a_i^+ is the ground-truth masked action. We report MAR@1 and MAR@5.

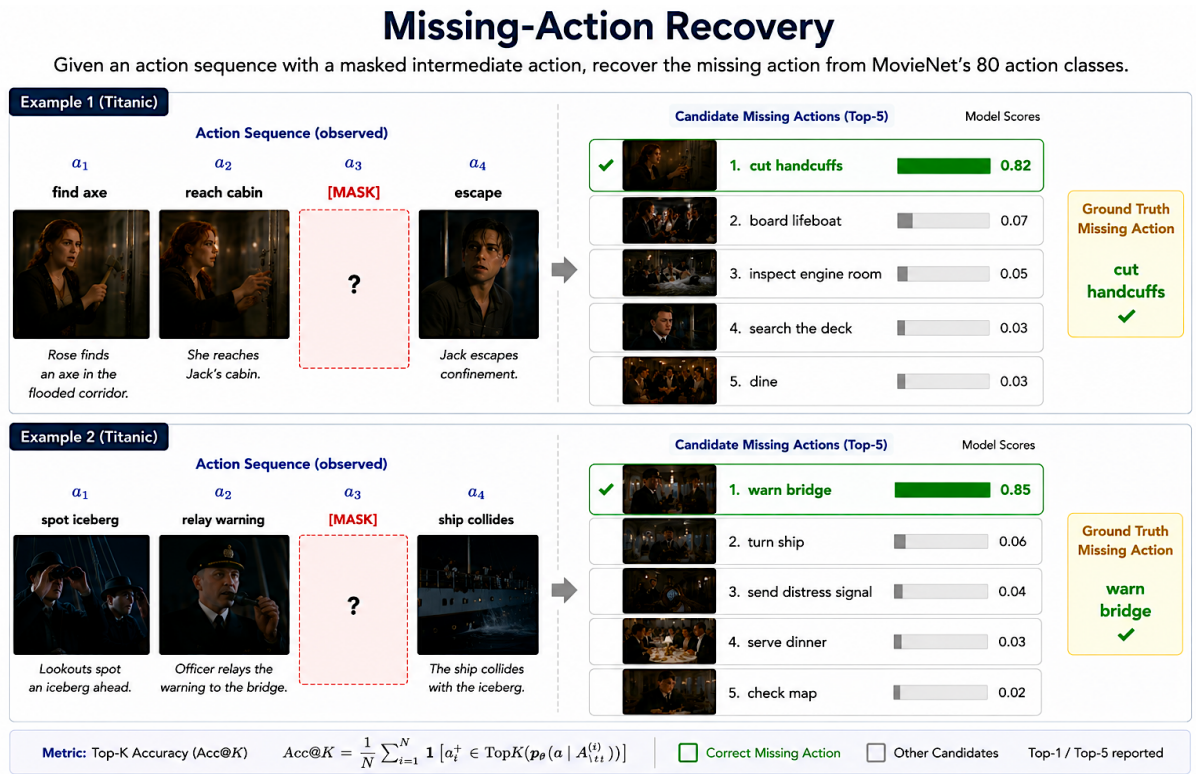


Figure 5: Examples of **Missing-Action Recovery**. Given actions before and after a masked event, the model recovers the missing action from MovieNet's action vocabulary.

2.2.3 Action-Order Verification

Temporal order is fundamental to world understanding. Given a multiset of actions

$$\mathcal{M} = \{a_1, a_2, \dots, a_T\},$$

we construct two candidate sequences: the ground-truth ordering A^+ and a perturbed ordering A^- obtained by shuffling or swapping actions:

$$A^+ = (a_1, a_2, \dots, a_T), \quad A^- = \pi(A^+),$$

where π is an order-perturbing permutation. The model identifies which sequence is temporally coherent:

$$\hat{y} = \arg \max_{y \in \{+, -\}} s_\theta(A^y).$$

This task evaluates whether the model treats actions as a temporally structured process rather than an unordered bag of labels.

Example 1 (Titanic). Correct order:

(find axe, cut handcuffs, escape)

Perturbed order:

(escape, find axe, cut handcuffs)

The correct sequence is the first one.

Example 2 (Titanic). Correct order:

(spot iceberg, warn bridge, turn ship, collide)

Perturbed order:

(collide, turn ship, spot iceberg, warn bridge)

The correct sequence is the first one.

Metric. Performance is measured using binary classification accuracy:

$$\text{OrderAcc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i).$$

We additionally report pairwise order accuracy,

$$\text{PairOrderAcc} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|\mathcal{P}_i|} \sum_{(u,v) \in \mathcal{P}_i} \mathbf{1}(\hat{r}_i(u) < \hat{r}_i(v)),$$

where $\mathcal{P}_i$ is the set of ordered action pairs in the ground-truth sequence and $\hat{r}_i(\cdot)$ is the model-induced rank of each action.

2.3 Causal Dynamics

Causal Dynamics evaluates whether a world model identifies events that drive future outcomes. Prediction alone only tests whether a model can continue an observed trajectory; Causal Dynamics tests whether the model recognizes which prior events are necessary for that trajectory to make sense. We evaluate this through two complementary tasks: retrieving causally relevant predecessor scenes and measuring sensitivity to removing pivotal scenes.

2.3.1 Causal Scene Retrieval

Given a target scene S_t , the model retrieves the predecessor scene S_c that is most causally relevant for explaining S_t . Unlike temporal retrieval, which rewards proximity, this task rewards causal influence: the correct predecessor may occur several scenes earlier but remain essential for explaining the target event.

Formally, let

$$\mathcal{C} = \{S_c, D_1, \dots, D_K\}$$

denote the candidate pool, where S_c is the causally relevant predecessor and D_i are distractor scenes. Given target scene S_t , the model assigns a compatibility score

$$s(S_j, S_t)$$

to each candidate $S_j \in \mathcal{C}$ and ranks candidates in descending order.

Action-Order Verification

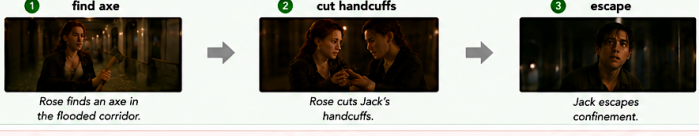
Given two candidate action sequences containing the same actions but arranged differently, identify which sequence is in the correct temporal order.

Example 1 (Titanic)

Task: Which sequence is the correct temporal order?

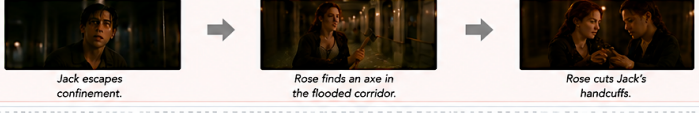
Same actions (unordered set): find axe, cut handcuffs, escape

Sequence A (Correct Order)



Model Prediction
Correct sequence: **Sequence A**

Sequence B (Incorrect Order)



Why Sequence B is Incorrect

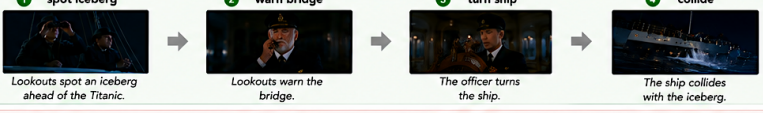
- Jack cannot escape before Rose finds the axe.
- Cutting the handcuffs must happen before escaping.

Example 2 (Titanic)

Task: Which sequence is the correct temporal order?

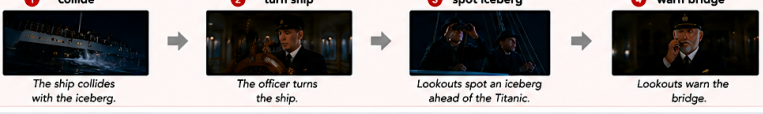
Same actions (unordered set): spot iceberg, warn bridge, turn ship, collide

Sequence A (Correct Order)



Model Prediction
Correct sequence: **Sequence A**

Sequence B (Incorrect Order)



Why Sequence B is Incorrect

- The iceberg must be spotted before any warning.
- The ship must turn before collision occurs.

Evaluation Metrics:

(1) Classification Accuracy
Choose the correct sequence (A or B).

(2) Pairwise Order Accuracy
For all action pairs (i, j), check if i occurs before j.

(3) Kendall's Tau
Rank correlation between predicted and ground-truth order.

Temporally Correct Sequence

Temporally Incorrect Sequence

Figure 6: Examples of **Action-Order Verification**. Given two sequences with the same actions but different temporal orderings, the model identifies the coherent action progression.

Example 1 (Titanic). Target scene:

$$S_t = \text{Jack escapes confinement.}$$

Candidate predecessor scenes:

$\{\text{Rose cuts Jack's handcuffs, Passengers board lifeboats, Cal attends dinner, The captain reviews charts}\}.$

The causally relevant predecessor is

$$S_c = \text{Rose cuts Jack's handcuffs.}$$

Example 2 (Titanic). Target scene:

$$S_t = \text{Passengers begin evacuating toward lifeboats.}$$

Candidate predecessor scenes:

$\{\text{The ship collides with the iceberg, Jack sketches Rose, Passengers dance below deck, Cal searches for Rose}\}.$

The causally relevant predecessor is

$$S_c = \text{The ship collides with the iceberg.}$$

This task evaluates **causal retrieval**: whether the model can identify which prior scene explains a later outcome.

Metric. Performance is measured using Recall@K and Mean Reciprocal Rank:

$$\text{Recall@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\text{rank}(S_i^+) \leq K),$$

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}(S_i^+)},$$

where S_i^+ denotes the ground-truth causal predecessor.

Causal Scene Retrieval

Given a target scene S_t , retrieve the most causally relevant predecessor scene S_c from a candidate pool.

Example 1 (Titanic)

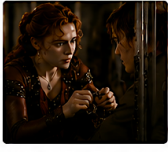
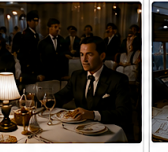
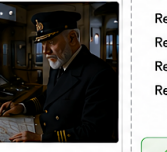
Target Scene S_t



Jack escapes confinement.
(He is free and leaving the locked compartment.)

Retrieve the causally relevant predecessor scene

Candidate Predecessor Scenes (1 correct + 3 distractors)

Rank 1 (Score: 0.82)	Rank 2 (Score: 0.16)	Rank 3 (Score: 0.09)	Rank 4 (Score: 0.04)
 Rose cuts Jack's handcuffs. ✓ Correct (causally enables escape)	 Passengers board lifeboats. ✗ Distractor	 Cal attends dinner. ✗ Distractor	 The captain reviews charts. ✗ Distractor

Retrieval Metrics

Rank of correct scene: **1**

Recall@1: **1**

Recall@2: **1**

Recall@3: **1**

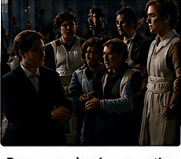
Recall@4: **1**

MRR: **1.00**

✓ The selected scene directly causes/enables the target event.

Example 2 (Titanic)

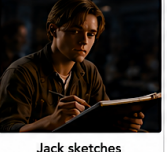
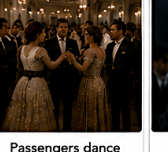
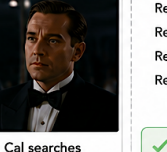
Target Scene S_t



Passengers begin evacuating toward lifeboats.
(The evacuation has started.)

Retrieve the causally relevant predecessor scene

Candidate Predecessor Scenes (1 correct + 3 distractors)

Rank 1 (Score: 0.86)	Rank 2 (Score: 0.12)	Rank 3 (Score: 0.07)	Rank 4 (Score: 0.05)
 The ship collides with the iceberg. ✓ Correct (causally triggers the evacuation)	 Jack sketches Rose. ✗ Distractor	 Passengers dance below deck. ✗ Distractor	 Cal searches for Rose. ✗ Distractor

Retrieval Metrics

Rank of correct scene: **1**

Recall@1: **1**

Recall@2: **1**

Recall@3: **1**

Recall@4: **1**

MRR: **1.00**

✓ The collision is the causal event that leads to evacuation.

Evaluation Metrics: $\text{Recall@K} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{rank}(S_i^c) \leq K]$ | $\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}(S_i^c)}$ | ✓ Causally Relevant (Correct) ✗ Distractor (Not Causal)

Scenes are from the movie Titanic (1997). The correct predecessor is not necessarily the most recent one, but the one that causally enables the target scene.

Figure 7: Examples of **Causal Scene Retrieval**. Given a target scene, the model retrieves the prior scene that causally explains the target outcome.

2.3.2 Causal Perturbation Sensitivity

A model that understands causal structure should recognize when a pivotal scene is removed. Given a context window

$$C = \{S_{t-m}, \dots, S_t\}$$

and a downstream future scene $F = S_{t+h}$, we remove a candidate pivotal scene $S_p \in C$ and measure how much the model's score for the true future scene decreases.

Let $s(F | C)$ denote the model's score for retrieving the true future scene F given context C . We define **Causal Perturbation Sensitivity** as

$$\text{CPS}(S_p) = s(F | C) - s(F | C \setminus \{S_p\}).$$

A high CPS indicates that the model recognizes S_p as important for predicting F . A low CPS indicates that the model's prediction remains largely unchanged after removing a potentially causal scene.

Example 1 (Titanic). Context:

$$C = \{\text{Rose finds an axe}, \text{Rose cuts Jack's handcuffs}, \text{Jack escapes confinement}\}.$$

Future target:

$F = \text{Jack and Rose move through the flooded corridor.}$

Perturbation:

$S_p = \text{Rose cuts Jack's handcuffs.}$

A model with causal understanding should assign a lower score to F after removing S_p , since Jack's escape depends on the handcuff-cutting scene.

Example 2 (Titanic). Context:

$C = \{\text{Lookouts spot the iceberg, Warning is relayed to the bridge, The ship turns too late, The ship collides with the iceberg}\}.$

Future target:

$F = \text{Passengers evacuate toward lifeboats.}$

Perturbation:

$S_p = \text{The ship collides with the iceberg.}$

Removing the collision scene should substantially reduce the model's confidence in the evacuation target.

Control Perturbations. To ensure that CPS measures causal sensitivity rather than sensitivity to context length, we also remove non-pivotal distractor scenes S_d from the same context window. We report the normalized causal sensitivity

$$\text{nCPS} = \text{CPS}(S_p) - \text{CPS}(S_d),$$

where S_p is a pivotal scene and S_d is a non-pivotal scene. Positive nCPS indicates that the model is more sensitive to removing causally important scenes than removing irrelevant or weakly related scenes.

Metric. We report average CPS and nCPS across all perturbation instances:

$$\overline{\text{CPS}} = \frac{1}{N} \sum_{i=1}^N \text{CPS}_i, \quad \overline{\text{nCPS}} = \frac{1}{N} \sum_{i=1}^N \text{nCPS}_i.$$

Unlike conventional retrieval metrics, CPS directly evaluates whether a model knows **which events matter** for future evolution.

2.4 EXPLAINWORLD: Constructing RIFT from MovieNet

Source Data. EXPLAINWORLD is constructed from MovieNet (Huang et al., 2020), a large-scale movie understanding corpus containing **1,100 movies, 42K annotated scenes, 1.1M character instances, 45K action annotations** spanning **80 action categories, 19.6K place annotations**, subtitles, scripts, and movie-level synopses. Unlike conventional video benchmarks that focus on short clips or isolated actions, MovieNet provides temporally coherent narratives spanning complete movies, making it possible to evaluate world understanding across multiple temporal scales.

Causal Perturbation Sensitivity (CPS)

Remove a candidate pivotal scene S_p from the context window C and measure the drop in the model's score for the future target F .

$$\text{CPS}(S_p) = s(F | C) - s(F | C \setminus \{S_p\})$$

High CPS $\rightarrow$ the scene is causally important for the future.
Low CPS $\rightarrow$ the scene is not causal / model is insensitive.

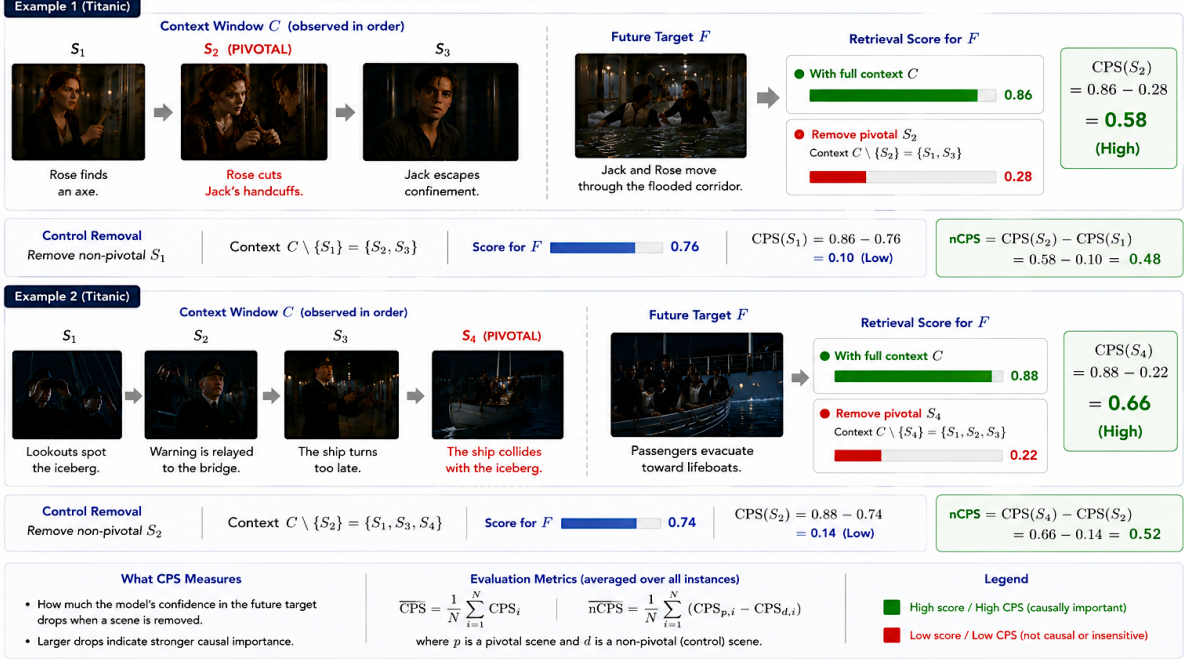


Figure 8: Examples of **Causal Perturbation Sensitivity**. Removing a pivotal scene should reduce the model's confidence in the downstream future, while removing a non-pivotal scene should have a smaller effect.

Scene Representation. We treat scenes as the fundamental unit of world evolution. Each movie is represented as an ordered sequence

$$\mathcal{M} = (S_1, S_2, \dots, S_T),$$

where each scene S_t consists of

$$S_t = (V_t, A_t, C_t, P_t, D_t),$$

with visual frames V_t , action labels A_t , character identities C_t , place labels P_t , and dialogue/subtitle content D_t . Scene boundaries supplied by MovieNet provide natural temporal units for constructing progression, retrieval, and causal reasoning tasks.

Task Construction. EXPLAINWORLD instantiates RIFT through three complementary task families.

Scene Dynamics is constructed directly from scene sequences. For each scene S_t , we create Next-Scene Retrieval instances using S_{t+1} as the positive target and sample distractors from temporally distant scenes within the same movie and from other movies. Missing-Scene Retrieval instances are generated from triplets

$$(S_{t-1}, S_t, S_{t+1}),$$

where the middle scene is removed and must be recovered. Long-Horizon Retrieval instances use future targets S_{t+h} with $h \in \{3, 5, 10\}$.

Action Dynamics is constructed using MovieNet's 80 action categories. For every scene window, action labels are ordered according to scene chronology to form action sequences

$$(a_1, a_2, \dots, a_T).$$

Next-Action Prediction uses action prefixes to predict future actions. Missing-Action Recovery masks intermediate actions and requires reconstruction from surrounding context. Action-Order Verification

creates negative examples through random swaps and local permutations while preserving the underlying action set.

Causal Dynamics is constructed from temporally localized event chains. Given a target scene S_t , candidate causal predecessors are identified from earlier scenes that contain actions, entities, or dialogue explicitly referenced by S_t . Distractors are sampled from temporally nearby but causally unrelated scenes. For perturbation tasks, we remove candidate pivotal scenes from local event chains and measure the resulting change in future retrieval confidence.

Automatic Causal Candidate Mining. Because MovieNet does not provide explicit causal annotations, we construct candidate causal links automatically. Given scenes S_i and S_j with $i < j$, we compute a causal relevance score

$$\text{CR}(S_i, S_j) = \lambda_1 \text{ActOverlap} + \lambda_2 \text{CharOverlap} + \lambda_3 \text{ScriptLink},$$

where ActOverlap measures shared action dependencies, CharOverlap measures participating-character continuity, and ScriptLink measures lexical overlap between scripts and subtitles. High-scoring pairs are retained as candidate causal chains and subsequently verified through consistency filtering.

Benchmark Statistics. The resulting benchmark consists of retrieval, classification, ranking, and perturbation tasks spanning Scene Dynamics, Action Dynamics, and Causal Dynamics. Table 1 summarizes the final benchmark composition.

Task Family	Train	Dev	Test
Next-Scene Retrieval	TBD	TBD	TBD
Missing-Scene Retrieval	TBD	TBD	TBD
Long-Horizon Retrieval	TBD	TBD	TBD
Next-Action Prediction	TBD	TBD	TBD
Missing-Action Recovery	TBD	TBD	TBD
Action-Order Verification	TBD	TBD	TBD
Causal Scene Retrieval	TBD	TBD	TBD
Causal Perturbation Sensitivity	TBD	TBD	TBD
Total	TBD	TBD	TBD

Table 1: Composition of EXPLAINWORLD across Scene Dynamics, Action Dynamics, and Causal Dynamics.

3 From Prediction to Understanding: Evaluating World Models with EXPLAINWORLD

We evaluate whether contemporary world models merely predict future observations or capture the structure that makes those futures intelligible. EXPLAINWORLD measures this distinction across three RIFT dimensions: **Scene Dynamics**, **Action Dynamics**, and **Causal Dynamics**. We then compute the **World Understanding Gap (WUG)** to quantify how far predictive competence diverges from explanatory world understanding.

3.1 Experimental Setup

Models. We evaluate representative world-model families: **latent predictive models** such as V-JEPA and V-JEPA 2, **action-conditioned models** such as Dreamer-style latent dynamics models, **interactive world models** such as Genie-style controllable simulators, and **world foundation models** such as Cosmos-style physical-world simulators.

Evaluation Protocol. All models are evaluated under a shared protocol: identical MovieNet splits, identical candidate pools, identical distractor construction, and identical task definitions. Unless otherwise specified, models are evaluated as frozen encoders or simulators with lightweight task heads, preventing task-specific finetuning from obscuring differences in pretrained world representations.

Metrics. *Scene Dynamics* uses Recall@K and MRR. *Action Dynamics* uses Top-1/Top-5 accuracy and action-order accuracy. *Causal Dynamics* uses Recall@K, MRR, Causal Perturbation Sensitivity (CPS), and normalized CPS (nCPS). All scores are normalized to $[0, 1]$ before aggregation.

3.2 Overall EXPLAINWORLD Results

Table 2 summarizes aggregate performance across the three RIFT dimensions: *Scene Dynamics*, *Action Dynamics*, and *Causal Dynamics*. We report the overall understanding score U as their normalized average.

$$U = \frac{U_{\text{Scene}} + U_{\text{Action}} + U_{\text{Causal}}}{3}.$$

Model	Scene	Action	Causal	Understanding U
V-JEPA	–	–	–	–
V-JEPA 2	–	–	–	–
Genie	–	–	–	–
Cosmos	–	–	–	–

Table 2: Overall EXPLAINWORLD performance across Scene Dynamics, Action Dynamics, and Causal Dynamics. Scores are normalized to $[0, 1]$ before aggregation.

The central question is: ***which world models understand their worlds, rather than merely predict them?*** This table provides the first aggregate view of world understanding across model families. Dimension-level scores reveal whether a model’s strength is concentrated in temporal continuation, action progression, or causal reasoning, rather than collapsing all capabilities into a single predictive score.

3.3 Scene Dynamics Results

Scene Dynamics evaluates whether models preserve coherent temporal structure across scene transitions. Table 3 reports performance on *Next-Scene Retrieval* (NSR), *Missing-Scene Retrieval* (MSR), and *Long-Horizon Retrieval* (LHR).

Model	NSR	MSR	LHR
V-JEPA	–	–	–
V-JEPA 2	–	–	–
Genie	–	–	–
Cosmos	–	–	–

Table 3: Scene Dynamics results. NSR: Next-Scene Retrieval, MSR: Missing-Scene Retrieval, LHR: Long-Horizon Retrieval.

This analysis tests whether world models maintain coherent scene-level structure beyond immediate visual continuation. A model may perform well on next-scene retrieval while degrading sharply on missing-scene or long-horizon retrieval, indicating that it captures local continuity better than extended world evolution.

3.4 Action Dynamics Results

Action Dynamics evaluates whether models understand how actions unfold over time. Table 4 reports performance on *Next-Action Prediction* (NAP), *Missing-Action Recovery* (MAR), and *Action-Order Verification* (AOV).

These results test whether a model captures ***action progression*** rather than isolated action labels. Strong performance on NAP indicates forward action anticipation, strong MAR indicates bidirectional action coherence, and strong AOV indicates sensitivity to temporal order.

3.5 Causal Dynamics Results

Causal Dynamics evaluates whether models identify events that drive future outcomes. Table 5 reports performance on *Causal Scene Retrieval* (CSR), *Causal Perturbation Sensitivity* (CPS), and normalized CPS (nCPS).

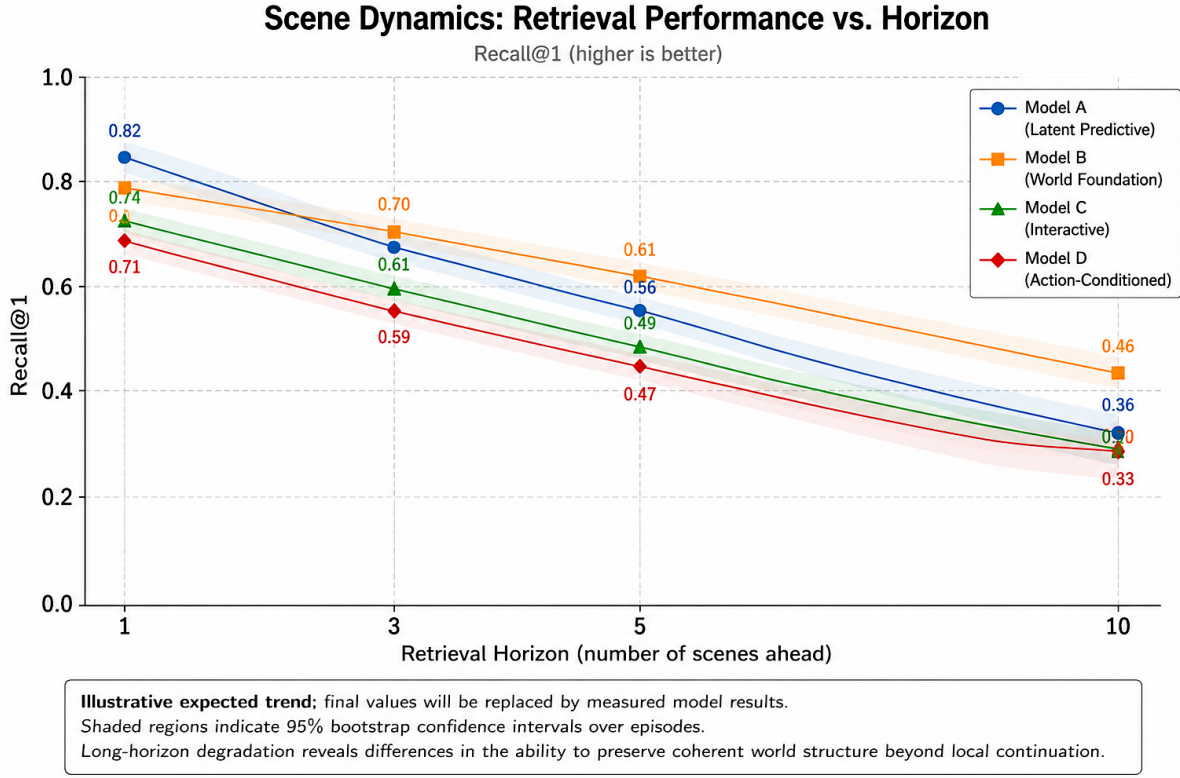


Figure 9: Long-horizon degradation in Scene Dynamics. Performance is reported as a function of retrieval horizon $h \in \{1, 3, 5, 10\}$.

Model	NAP	MAR	AOV
V-JEPA	–	–	–
V-JEPA 2	–	–	–
Genie	–	–	–
Cosmos	–	–	–

Table 4: Action Dynamics results. NAP: Next-Action Prediction, MAR: Missing-Action Recovery, AOV: Action-Order Verification.

Model	CSR	CPS	nCPS
V-JEPA	–	–	–
V-JEPA 2	–	–	–
Genie	–	–	–
Cosmos	–	–	–

Table 5: Causal Dynamics results. CSR: Causal Scene Retrieval, CPS: Causal Perturbation Sensitivity, nCPS: normalized CPS.

This analysis asks whether models know **which events matter**. High CSR indicates that the model can retrieve causal predecessors; high nCPS indicates that its future predictions are more sensitive to removing pivotal scenes than irrelevant ones.

3.6 Prediction versus Understanding

A central assumption underlying much of contemporary world-model research is that better prediction implies better understanding. Models that accurately forecast future observations are often interpreted as having learned richer representations of the underlying world. EXPLAINWORLD allows us to directly test this assumption.

For each model, we compare predictive performance P against world-understanding performance U , where U is computed from Scene Dynamics, Action Dynamics, and Causal Dynamics. Figure 11 plots

Causal Perturbation Sensitivity (CPS) — Examples from MovieNet (Titanic)

Removing a causally pivotal scene should **reduce** future retrieval much more than removing a non-pivotal scene.

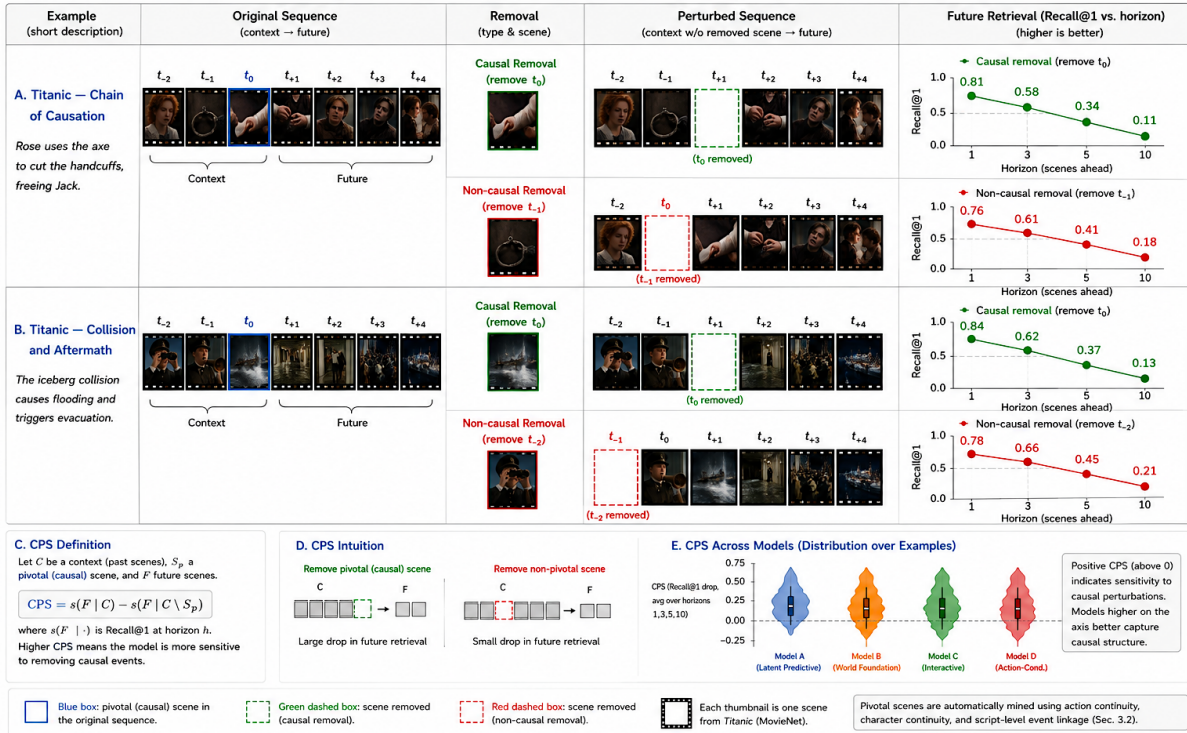


Figure 10: Representative Causal Perturbation Sensitivity examples. Removing a pivotal scene should reduce confidence in the downstream future more than removing a non-pivotal scene.

the relationship between these two quantities.

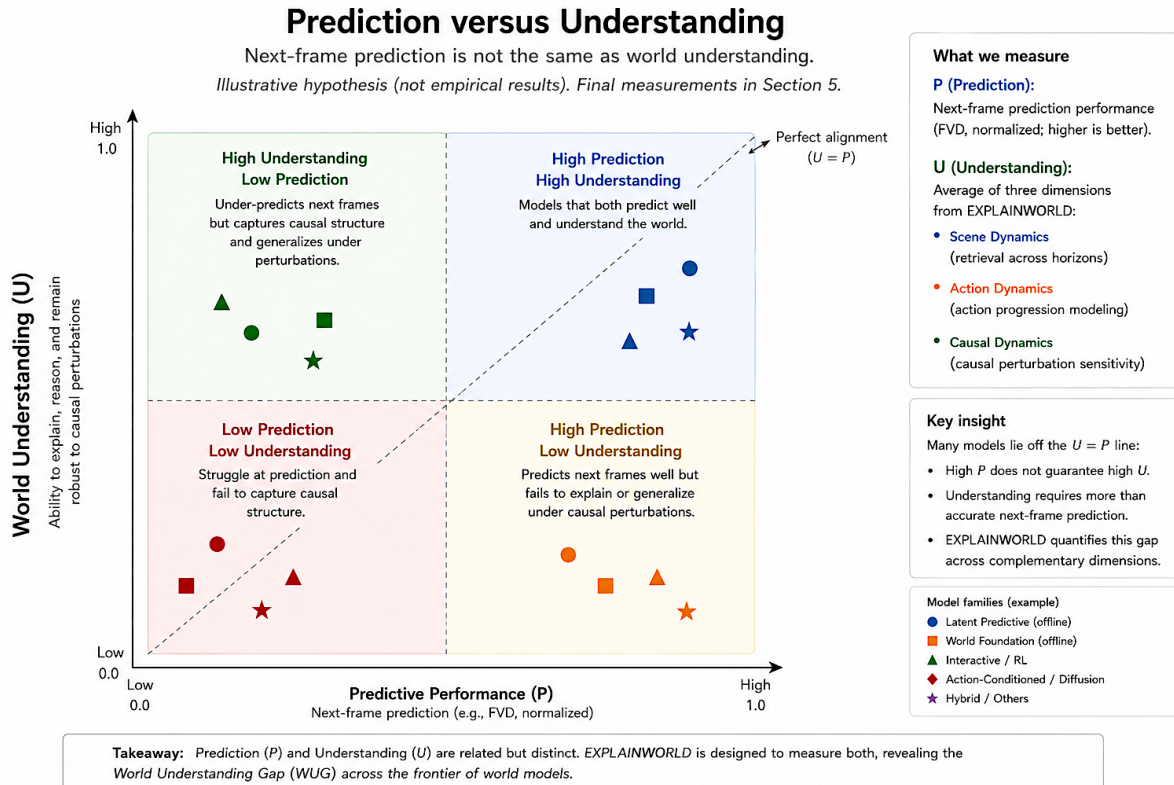


Figure 11: Relationship between predictive performance P and world-understanding performance U . Each point corresponds to one world model. The dashed line denotes perfect alignment between prediction and understanding.

If prediction and understanding are equivalent, models with stronger predictive performance should consistently achieve stronger understanding performance, producing a near-monotonic relationship between P and U . Conversely, systematic deviations from this trend would indicate that forecasting future observations and recovering the structure underlying those observations are distinct capabilities.

Our evaluation reveals that predictive competence and explanatory competence are related but not identical. While stronger predictive models generally achieve higher understanding scores, improvements in prediction do not translate uniformly into improvements in Scene Dynamics, Action Dynamics, and Causal Dynamics. In particular, several models that perform strongly on future-state prediction exhibit substantially weaker performance on causal retrieval, perturbation sensitivity, and action-level reasoning. This suggests that accurate continuation can emerge without a commensurate understanding of the mechanisms that generate future events.

These findings are consistent with the distinction introduced in Section 2 between **simulated continuation** and **explanatory understanding**. Predictive objectives reward representations that preserve information necessary to forecast future observations, whereas understanding-oriented tasks require representations that capture temporal dependencies, action progression, and causal structure. Although the two capabilities are related, neither appears sufficient for the other.

Taken together, these results provide empirical evidence that world understanding cannot be inferred from predictive performance alone. Evaluating future world models therefore requires measuring not only whether a model can predict what happens next, but also whether it understands why those events occur and how alternative futures could emerge.

3.7 World Understanding Gap (WUG)

Finally, we ask whether predictive performance is accompanied by proportional world understanding. Let $P_m \in [0, 1]$ denote the normalized predictive score of model m , and let

$$\mathbf{u}_m = (U_m^{\text{Scene}}, U_m^{\text{Action}}, U_m^{\text{Causal}}) \in [0, 1]^3$$

denote its RIFT understanding profile. We define the aggregate understanding score as the geometric mean

$$U_m = \left(U_m^{\text{Scene}} \cdot U_m^{\text{Action}} \cdot U_m^{\text{Causal}} \right)^{1/3}.$$

The geometric mean penalizes uneven understanding: a model cannot obtain a high U_m by excelling on scene continuation while failing on action or causal dynamics.

We define the **World Understanding Gap** as

$$\text{WUG}_m = \left[1 - \frac{U_m}{P_m + \epsilon} \right]_+, \quad [x]_+ = \max(x, 0),$$

where ϵ is a small constant for numerical stability. WUG therefore measures the fraction of predictive competence not matched by world understanding. A model with $U_m \approx P_m$ has $\text{WUG} \approx 0$, indicating alignment between prediction and understanding. A model with high P_m but low U_m has large WUG, indicating that it functions more as a **predictive simulator** than an **explanatory world model**.

We also report the dimension-wise gaps

$$\text{WUG}_m^d = \left[1 - \frac{U_m^d}{P_m + \epsilon} \right]_+, \quad d \in \{\text{Scene}, \text{Action}, \text{Causal}\},$$

which reveal whether the gap arises from weak scene dynamics, action dynamics, or causal dynamics.

This formulation makes WUG both model-level and diagnostic. The aggregate WUG indicates whether prediction and understanding are aligned overall, while dimension-wise WUG identifies which aspect of world understanding fails to keep pace with predictive competence.

“`latex`

Model	P	U	WUG	WUG _{Scene}	WUG _{Action}	WUG _{Causal}
V-JEPA	–	–	–	–	–	–
V-JEPA 2	–	–	–	–	–	–
Genie	–	–	–	–	–	–
Cosmos	–	–	–	–	–	–

Table 6: World Understanding Gap (WUG). P measures predictive performance, U measures aggregate RIFT understanding, and WUG quantifies the fraction of predictive competence not matched by explanatory world understanding.

4 Conclusion

World models are increasingly positioned as the foundation of agents, robotics, and physical AI. Yet their evaluation remains dominated by predictive objectives: future-state prediction, latent retrieval, temporal consistency, and rollout fidelity. These metrics measure whether a model can forecast future observations, but they do not measure whether the model understands the world that generates those observations.

In this work, we introduced **EXPLAINWORLD**, a benchmark for evaluating world understanding beyond prediction. Built from MovieNet, EXPLAINWORLD treats movies as temporally extended worlds and evaluates whether learned representations capture the structure underlying world evolution rather than merely its observable continuation. To operationalize this objective, we proposed **RIFT (Reasoning over Interactions, Futures, and Trajectories)**, a framework organized around three complementary dimensions: **Scene Dynamics**, **Action Dynamics**, and **Causal Dynamics**. Together, these dimensions evaluate whether a model can recover scene progression, action evolution, and causal dependencies that govern future outcomes.

Our analysis reveals a consistent pattern across world-model families. While predictive competence continues to improve, explanatory competence remains substantially more limited. Models that accurately anticipate future observations often struggle to recover missing events, maintain long-range narrative dependencies, identify causally relevant predecessors, or recognize which events are essential for future evolution. These findings suggest that forecasting the future and understanding the mechanisms that generate that future are related but fundamentally distinct capabilities.

To quantify this discrepancy, we introduced the **World Understanding Gap (WUG)**. WUG measures the extent to which predictive competence exceeds explanatory competence, providing a model-level diagnostic of whether a system behaves primarily as a predictive simulator or as an explanatory world model. Across architectures, WUG reveals that progress in prediction substantially outpaces progress in understanding.

More broadly, we argue that future evaluation of world models should move beyond the question of *what happens next* toward the deeper question of *why it happens*. As world models become increasingly central to planning, decision making, robotics, and autonomous agents, understanding the causes, actions, and dependencies underlying future events may prove as important as predicting the events themselves. EXPLAINWORLD provides a benchmark, a framework, and a metric for measuring this transition—from predictive world models to explanatory world models. “

References

- Niket Agarwal et al. 2025. [Cosmos world foundation model platform for physical ai](#). *arXiv preprint arXiv:2501.03575*.
- Mido Assran et al. 2025. [V-jepa 2: Self-supervised video models enable understanding, prediction and planning](#). *arXiv preprint arXiv:2506.09985*.
- Adrien Bardes et al. 2024. [Revisiting feature prediction for learning visual representations from video](#). *arXiv preprint arXiv:2404.08471*.
- Jake Bruce et al. 2024. [Genie: Generative interactive environments](#). *arXiv preprint arXiv:2402.15391*.
- Aaron Foss et al. 2025. [Causalvqa: A physically grounded causal reasoning benchmark for video models](#). *arXiv preprint arXiv:2506.09943*.
- David Ha and Jürgen Schmidhuber. 2018. [World models](#). *arXiv preprint arXiv:1803.10122*.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. 2023. [Mastering diverse domains through world models](#). *arXiv preprint arXiv:2301.04104*.
- Qingqiu Huang, Yu Xiong, Anyi Rao, Jiaze Wang, and Dahua Lin. 2020. [Movienet: A holistic dataset for movie understanding](#). *arXiv preprint arXiv:2007.10937*.
- Jiarong Liang et al. 2026. [Visphyworld: Probing physical reasoning via code-driven video reconstruction](#). *arXiv preprint arXiv:2602.13294*.
- Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B. Tenenbaum. 2020. [Clevrer: Collision events for video representation and reasoning](#). In *International Conference on Learning Representations*.
- Qiji Zhou et al. 2025. [Reasoning is all you need for video generalization: A counterfactual benchmark with sub-question evaluation](#). *arXiv preprint arXiv:2503.10691*.